

The Role of Human Engineers and Mathematicians in the Age of AI

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Abstract. AI is here: more people than ever are using AI and machine learning in their day-to-day operations, with Claude writing more code than humans and with deep neural networks becoming a go-to method for many applications. This raises the big question: What role will mathematicians and engineers play in future workflows given this AI-based future? In this talk we will explore many open areas and unanswered questions arising from the intersection of scientific machine learning and agentic AI. Specifically, we will examine the role of mathematical systems for code generation (formalization of automatic differentiation and related techniques), the development of new languages for static analysis and formal verification to improve the accuracy of agentic AI coding systems for physical modeling, and the integration of symbolic-numeric computing into scientific machine learning APIs. A showcase of how industrial processes are changing will be mixed with discussions of academic research, showing the full pipeline of how hypotheses are being transitioned into real-world applications for early adoptors of AI integrated technologies.